

Probing Physical Affordance Understanding

Benchmark for evaluation of CLIP vs JEPA models

Motivation: VL-JEPA claims to understand world knowledge and by shifting supervision from discrete token space to continuous semantic embedding space. This experiment is designed to test how well the joint embedding predictive architectures encode the affordance knowledge of the objects. It is able to capture “What is this?”, but can the model answer “What can I do with this?”. The motivation is to evaluate how well these architectures generalise to real-world visual data, where object states, part visibility, and physical context vary naturally rather than being controlled.

Dataset: [PACO-LVIS dataset](#)

PACO: Parts and Attributes of Common Objects.

It spans 75 object categories, 456 objectpart categories and 55 attributes across image (LVIS) and video (Ego4D) datasets. We use this dataset as, it captures images that we in everyday life, blenders, cups, bottles knives in different background and setting which enables us to evaluate the affordance understanding in every setting.

A blender with white base



A metal mug with black handle



Pipeline:



First, perform a pilot study: 20 images sampled, then apply it to the image corpus

1. Dataset as explained above
2. We use Qwen 2.5-VL-7B for caption generation for the most probable and the negative captions. Example JSON here [1]
3. We then perform adversarial filtering on these captions with CLIP zero-shot to remove the easy affordance captions (can make this process iterative to generate hard captions).
4. Inference: Are the representations geometrically aligned for affordances, without adding training parameters. We are trying to understand what the model already knows. Visualize using PCA in 2D, whether the correct pairs cluster together
 - a. **CLIP**
Compute cosine similarity for both captions against the image embedding. Whichever scores higher is the model's choice. Simple argmax over two scores.
 - b. **VL-JEPA**
Computing distance between the image and the caption embedding, the closer caption in the latent space is the model's choice.
5. Evaluation metrics:
 - a. Binary Accuracy: % most probable vs negative
 - b. Confidence gap: to evaluate if the model projections were really far apart, and the difference in similarity is not a very tiny margin.
 - c. Failure breakdown: evaluating where the model chose wrong. (% spatial / attribute/fluency failures)

Improving: On failure analysis, the model can be finetuned to address the failures. This could then determine whether the dominant failure type is addressable through a targeted contrastive signal or reflects a deeper representational limitation of the frozen architecture.

Example JSON:

```

{
  "objects": [
    {
      "object": "bottle with visible cap",
      "affordance_tiers": {
        "most_probable": [
          "Grip the bottle body to carry it securely.",
          "Hold the bottle side to steady it upright.",
          "Twist the cap to open the bottle."
        ]
      }
    },
  ],
}
```

```

    "negative": [
      "Unscrew the cap to access the contents.",
      "Twist the cap to refill the bottle."
    ]
  },
  {
    "object": "glass bowl",
    "affordance_tiers": {
      "most_probable": [
        "Hold the bowl rim to lift it carefully.",
        "Grip the bowl side to move it.",
        "Steady the bowl base to prevent sliding."
      ],
      "negative": [
        "Stack the bowl on a matching bowl.",
        "Pour the bowl contents into another container."
      ]
    }
  },
  {
    "object": "container with lid",
    "affordance_tiers": {
      "most_probable": [
        "Twist the lid to open the container.",
        "Grip the jar body to hold it steady.",
        "Press the lid down to close it."
      ],
      "negative": [
        "Pour contents from the opened container.",
        "Reach into the container to grab contents."
      ]
    }
  },
  {
    "object": "partially open drawer",
    "affordance_tiers": {
      "most_probable": [
        "Pull the drawer front to access contents.",
        "Push the drawer front to close it.",
        "Grip the handle to slide the drawer."
      ],
      "negative": [
        "Pull the drawer farther to access contents."
      ]
    }
  }
}

```

"Slide the drawer outward to inspect items."

```
}  
  ]  
    }  
  }  
}
```